

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

ME 19-205-0403 MECHATRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: **Understand** the basics of sensors and transducers and different mechatronics systems.

CO2: **Model** first and second order systems using block diagrams and governing equations.

CO3: **Analyse** first and second order systems for their time and frequency domain responses and stability.

CO4: **Design** mechatronics system for simple application.

CO5: **Use** software tools for modelling and simulation of mechatronic systems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

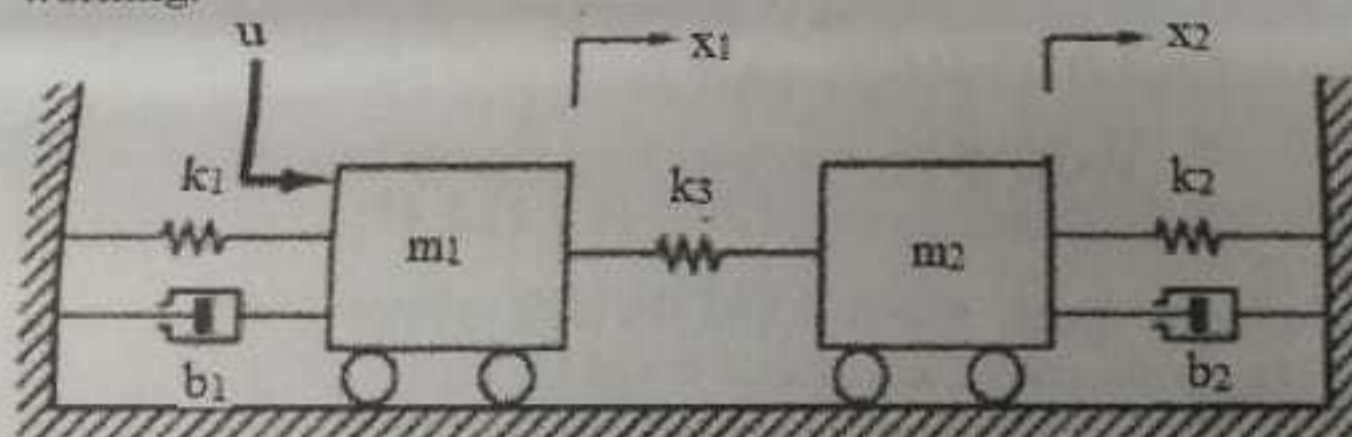
L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

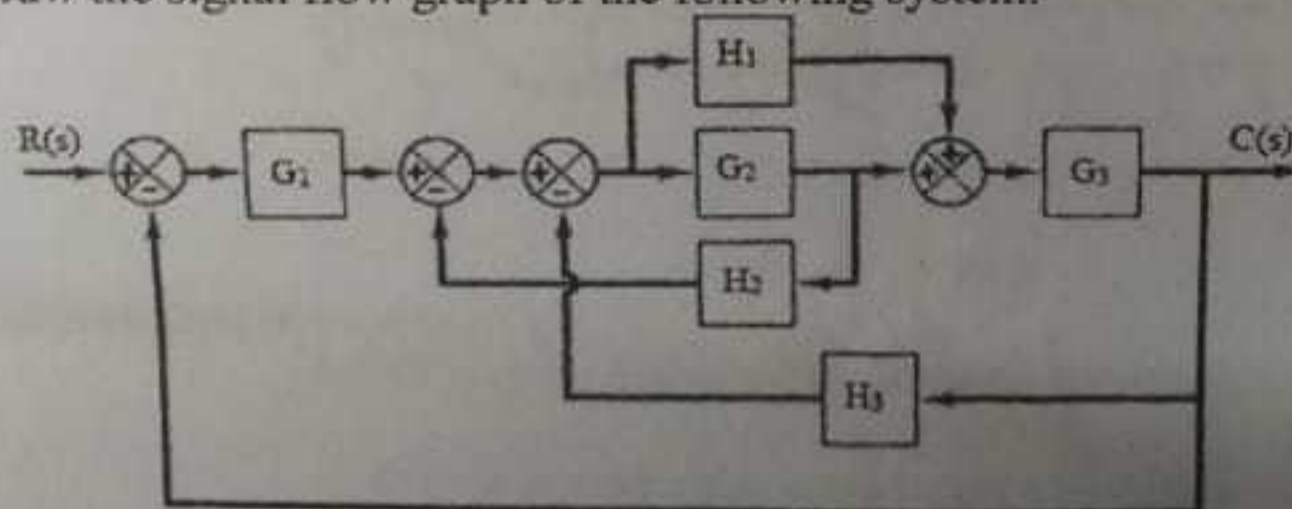
(Answer *ALL* questions)

- | I. | (8 × 3 = 24) | Marks | BL | CO | PO |
|--|--------------|-------|----|-----------------|----|
| (a) What is a Ratchet and pawl mechanism? Explain its working principle. | 3 | L1 | 1 | 1,2,3,4, 8,12 | |
| (b) Draw the symbol of a 5 port 2 position direction control valve which is actuated by solenoids on both sides and explain its working. | 3 | L2 | 1 | 1,2,3,4, 8,12 | |
| (c) | 3 | L3 | 2 | 1,2,3,4, 5,8,12 | |



For the mechanical system shown above, determine the force-voltage analogous system.

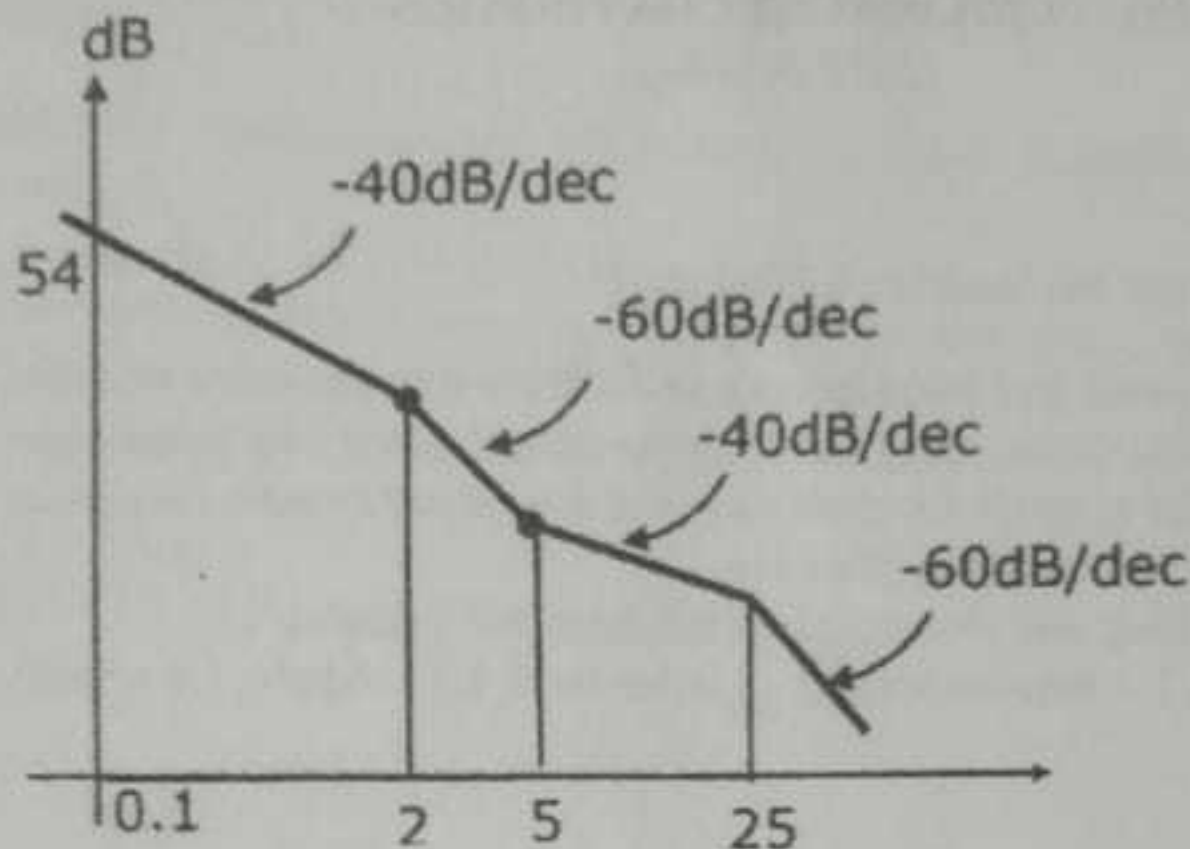
- | | | | | |
|---|---|----|---|-----------------|
| (d) Draw the signal flow graph of the following system. | 3 | L3 | 2 | 1,2,3,4, 5,8,12 |
|---|---|----|---|-----------------|



(P.T.O.)

BTS-IV(R/S)-06-22-0806

- (e) The asymptotic approximation of $20 \log(|G(j\omega)|)$ versus ω of a minimum phase system with real poles and one zero is shown below. Determine the transfer function of the system.



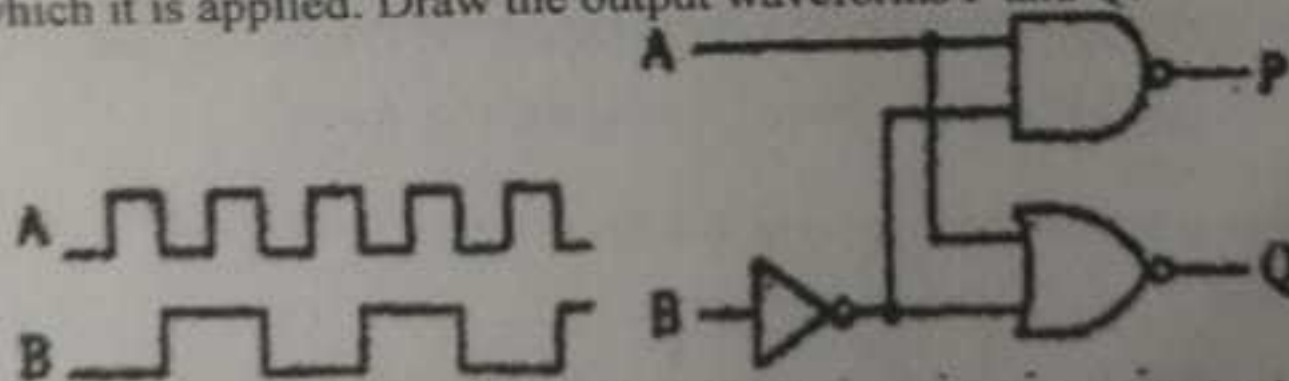
- (f) The open loop transfer function of a unity negative feed system is $G(s) = \frac{K}{s^2 + 4s - 5}$. Find the range of K for which the system will be stable.
- (g) Differentiate between a Micoprocessor and a Microcontroller.
- (h) Represent the following Boolean algebra equations in a PLC logic ladder diagram.

$$Z = A.\bar{B}.C + \bar{D} + E$$

PART B

(4 × 12 = 48)

- II. (a) The figure shows the input signals A and B and the gate system to which it is applied. Draw the output waveforms P and Q.



- (b) Explain the woking principle of an AC stepper motor and obtain the transfer function.

OR

- III. (a) List out the displacement sensors and explain the working principle of LVDT.
- (b) A 24 V lamp is to switched ON when a push button switch is pressed. It has to remain ON even after the push button is released. Design a circuit for this using a 24 V, 2-pole relay.

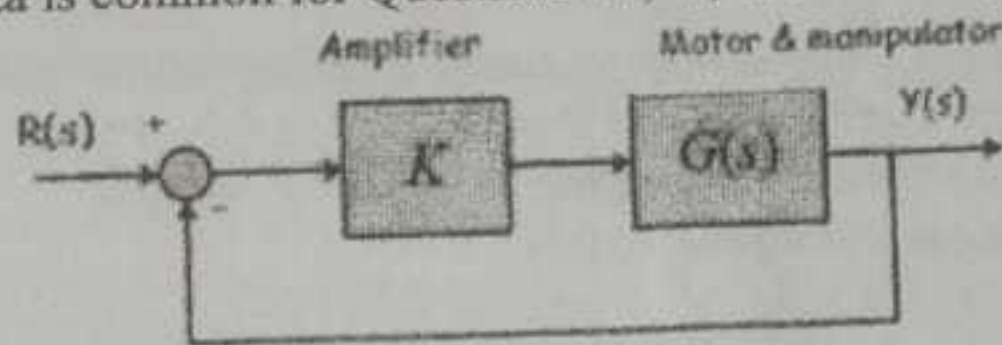
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- IV. The use of laser for surgery requires high accuracy for the position and velocity responses (or step and ramp inputs). The figure below shows the block diagram of a laser surgery system which uses a dc motor and a manipulator for the laser. The transfer function for the

$$\text{motor and manipulator is } G(s) = \frac{K}{s(s+10)(s+5)}$$

(This data is common for Questions IV, V, VI, and VII)



- (i) What is the range of values for K for which the system will be stable?
- (ii) Obtain the state space representation of the system for K=50.

6 L4 3 1,2,3,4, 5,8,12

6

OR

- V. (i) Referring to the system given in Question IV, what should be the amplifier gain K so that the steady state error for a ramp input $r(t) = At$ (where $A = 1 \text{ mm/s}$) should be less than or equal to 0.1 mm while remaining in stable condition?

4 L4 3 1,2,3,4, 5,8,12

- (ii) For that value of K found above, find the closed loop poles of the system.

2

- (iii) Can the system be approximated as a second order system? Justify your answer.

2

- (iv) If yes, determine the percentage overshoot and the 2% settling time of the system.

4

- VI. (i) Plot the root locus of the system given in Question IV.
- (ii) For an amplifier gain K=100, find the closed loop poles.
- (iii) Determine the transfer function of the PD controller for the above if the percentage overshoot is to be limited to 25% without any change in the settling time.

6 L4 3 1,2,3,4, 5,8,12

2

4 L5 4

OR

- VII. (i) Draw the frequency response curves (Bode diagram) for the system given in Question IV for an amplifier gain K=100
- (ii) Identify the gain margin and phase margin. Can you comment on the stability of the system from the margins?
- (iii) How much change can be made to the amplifier gain before the system becomes unstable?
- (iv) Determine the value of K_v from the Bode diagram and evaluate the velocity error constant.

6 L4 3 1,2,3,4, 5,8,12

2 L5 4

2

2

(P.T.O.)